Dynamos at ImageEval 2026 Shared Tasks: Metric-Aligned Selection and Cross-Family Ensembling for Vision-Language Hallucination Detection in English and Arabic

Muhammad Yamin
University of Girona
u6110888@udg.edu

Sheikh Abdul Wahid
Universidade do Algarve
a91643@ualg.pt

Ali Arshad
Skolkovo Institute of Science and Technology
Ali.Arshad@skoltech.ru

Abstract

We describe the Dynamos submission to ImageEval 2026 Task 1b, image-grounded hallucination detection: given an image and three statements, exactly one of which is grounded, a system must label each true or false. The official ranking metric, Contrastive Instability (CI, lower is better), reduces exactly to one minus the accuracy of picking the grounded statement once a system always outputs a single true label, which removes an entire failure mode (marking zero or several statements true) at no cost. Building on this reduction, we score each statement with an open vision-language model and always output the highest-scoring one as true. Starting from zero-shot Qwen2.5-VL, we low-rank fine-tune the model with a selection objective, train multiple random-seed adapters, and average their scores. Our main finding is that the strongest single model is not the strongest ensemble member: a 32B adapter trained on more data performs substantially better on the labelled `devtest` split but generalises worse than smaller 7B adapters to the blind test set, yet combining the two model families improves over either pure-family ensemble on both languages, despite the 32B family generalising worse on its own. Our final submitted systems reach CI 0.033 in English (2nd of 8 teams on the official leaderboard) and 0.056 in Modern Standard Arabic (3rd of 6 teams), both well below our zero-shot baselines of 0.058 (English) and 0.110 (MSA).

1 Introduction

ImageEval 2026 Task 1b evaluates whether vision-language systems can tell a statement that is actually supported by an image apart from a culturally plausible but hallucinated one (Abdaljalil et al., 2026). Each item pairs an image with three statements; exactly one is grounded, and the other two describe things that are locally plausible for the depicted culture or country but not shown in the image, in the spirit of the counterfactual hallucination setting studied by Mousi et al. (2026b) and the culturally grounded visual question answering data of Alam et al. (2025). We participated in both language tracks, English and Modern Standard Arabic (MSA), on Task 1b only.[1]

Our first contribution is not a model but a reading of the metric. The official ranking score, Contrastive Instability (CI), is computed from whether an item has at least one correct label and, among such items, whether all three are correct. We show that a system which always outputs exactly one `true` label removes the "zero correct" and "contradictory-pair" failure modes entirely, at which point CI collapses to one minus the accuracy of picking the grounded statement: the task becomes a three-way selection problem. This mirrors a lesson from our concurrent SHROOM-Visions submission, that reading a metric's exact definition and matching a system's output format to it is worth as much as model capacity.

Our second contribution is an empirical finding about ensembling across model scales under a tight compute and time budget. We fine-tune Qwen2.5-VL 7B and 32B (Qwen Team, 2025) with low-rank adapters (Hu et al., 2022; Dettmers et al., 2023) on the selection framing, training several 7B seeds on the training split alone and 32B adapters on the training split plus the labelled development split, to test whether the larger model could benefit from the additional 500 labelled examples. On the labelled `devtest` split the fine-tuned 32B is comfortably the best single model in both languages. On the blind test set the pattern reverses in English: an ensemble of four 32B adapters scores *worse* than an ensemble of three to five 7B adapters, even though the 32B adapters looked stronger on `devtest`. We hypothesise that

[1] Code: https://github.com/myamin66/ArabicNLPTask1B

Proceedings of The Fourth Arabic Natural Language Processing Conference (ArabicNLP 2026) - Volume 2 Shared Tasks, pages 681–688
October 28-29, 2026 ©2026 Association for Computational Linguistics

the higher-capacity 32B adapters are more sensitive to their training distribution than the 7B adapters, rather than genuinely better once evaluated on unseen data. Despite this, combining the two model families improves over either pure-family ensemble on both languages, which suggests that they make different errors even when one generalises worse alone.

Our final submitted systems reach blind-`test` CI of 0.033 (English) and 0.056 (MSA). For scale, our own Qwen2.5-VL-7B zero-shot selection scores 0.058 / 0.110 on `test`, and the organisers' released Qwen2.5-VL-3B baseline scores 0.313 / 0.490 (on the `devtest` split, and using independent per-statement True/False rather than our one-true selection, so it is a loose reference rather than a controlled comparison). On the official final leaderboard we place 2nd of 8 teams in English (top team: 0.029) and 3rd of 6 teams in MSA (top team: 0.036), within 0.004–0.020 CI of first place in both tracks.

2 Related Work

Task 1b (Abdaljalil et al., 2026) builds on the AynVQA/OASIS line of culturally grounded Arabic multimodal evaluation (Alam et al., 2025) and on the counterfactual-hallucination construction of Mousi et al. (2026b), where each item pairs a grounded statement with plausible-but-unsupported distractors; the sibling spoken track shows the same instability under cross-modal input (Mousi et al., 2026a). Deciding whether a vision-language model's claim is supported by an image is the multimodal counterpart of text hallucination detection studied in the SHROOM / Mu-SHROOM series. Our concurrent SHROOM-Visions submission (Yamin et al., 2026; Vázquez et al., 2026) addresses the character-level vision variant and is the source of two choices we reuse here: the LoRA gradient-checkpointing fix and the validation-to-test generalisation caution.

Methodologically, most Task 1b systems fine-tune a vision-language model and read a generated answer. Our angle differs in two ways: we read the answer from the model's next-token logits as a *selection* problem rather than generating it, which is exact and cheap under the metric reduction above; and we study *cross-family* (7B/32B) ensembling, finding that a model that generalises worse alone still helps as an ensemble member. Because MSA is our harder track, we also relate to work comparing native- and non-native-language prompting for Arabic, which finds that non-native (English) prompts are often competitive or better (Kmainasi et al., 2024).

3 Background

Task and data. Task 1b (AynVQA) provides, per item, an image and three natural-language statements about it; exactly one is grounded and the other two are hallucinated but culturally plausible, following the counterfactual-hallucination construction of Mousi et al. (2026b); the sibling spoken-question track (Task 1a) is itself subject to related cross-modal instability (Mousi et al., 2026a), underlining that grounding checks like ours are needed across input modalities, not only text prompts. Systems output a `true`/`false` label per statement. Data is released per language (English, MSA) with splits `train` (3,000 labelled items), `dev` (500, labelled), `devtest` (500, released with labels partway through the test phase), and the final blind `test` (1,000 items, scored on a hidden leaderboard during the competition). We participated in both language tracks of Task 1b only.

Metric. The organisers' scorer computes Contrastive Instability (CI, lower is better) alongside combined accuracy (all three labels correct), a contradictory-false-hallucination rate (CFHR), and two secondary accuracies. CI is defined only over the set S of items with at least one correct label, as the fraction of S that is not fully correct: $\text{CI} = |\{i \in S : i \text{ not all-correct}\}|/|S|$.

Derivation. The data guarantees exactly one grounded statement per item. If a system always emits one `true` (its guess) and two `false`, then when the guess is right all three labels are correct, and when it is wrong exactly one is correct (both `false` labels on the truly-false statements are right; the grounded and the falsely-chosen statement are each wrong). So every item has ≥ 1 correct label ($|S| = N$), and "not all-correct" coincides with a wrong guess, giving $\text{CI} = 1 - a$ with a the grounded-selection accuracy; $\text{CFHR} = 0$ because two `true` labels never occur (confirmed empirically in every run). The task thus reduces to three-way selection. This holds only because the data has exactly one grounded statement: with two or more, a single `true` label would leave some items with zero correct labels and $\text{CI} = 1 - a$ would fail. Our system relies on this guarantee.

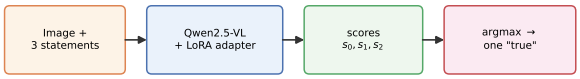


Figure 1: Inference pipeline for one adapter. The image and all three statements are fed to a LoRA/QLoRA fine-tuned Qwen2.5-VL in the *joint* framing; we read the model's first-token logits over the digits `0`/`1`/`2`, giving one score per statement, and emit the argmax as `true` and the other two as `false` (no text is generated). The final systems repeat this for several 7B and 32B adapters and average the softmax-normalised score vectors before the argmax (Section 4); the number and weighting of adapters per language are given in the Ensembling paragraph.

4 System Overview

Scoring a statement. We use Qwen2.5-VL (Qwen Team, 2025) as a scorer rather than a free-form generator. Two prompting framings are used throughout. The *per-statement* framing shows the image and one statement and reads the next-token log-probability of `True` minus `False` as that statement's grounded-ness score (summing sub-word-tokenisation variants of each word via log-sum-exp). The *joint* framing shows the image and all three statements together, numbered 0–2, and reads the first-token log-probabilities over the digits `0`, `1`, `2` as the three scores in one forward pass. In both framings the output is exactly one `true` label, the argmax, per Section 3.

Zero-shot baseline. Qwen2.5-VL-7B and -32B are scored zero-shot with the per-statement framing. 7B outperforms 32B in English (dev CI 0.054 vs. 0.068) while 32B is marginally better in MSA (0.082 vs. 0.086); 4-bit quantisation of the 32B consistently hurts and is not used zero-shot.

Fine-tuning. We low-rank fine-tune Qwen2.5-VL with the joint framing as supervised selection: the target is the digit of the grounded statement, and the loss is masked to that single output token. LoRA adapters (rank 16, $\alpha = 32$, dropout 0.05) are placed on all attention and MLP projections; the 32B backbone is loaded in 4-bit NF4 for a QLoRA-style adaptation (Dettmers et al., 2023) to fit one 80 GB accelerator. Training reuses the gradient-checkpointing fix (`use_reentrant=False` with `enable_input_require_grads`) from our SHROOM-Visions system (Yamin et al., 2026), without which LoRA parameters silently receive no gradient under checkpointing. Both languages are pooled into one multilingual training set. Full hyperparameters are in Appendix A.

Data allocation and its consequence. 7B adapters are trained on `train` only (multiple random seeds); 32B adapters are trained on `train+dev`, reasoning that the larger model could use the extra 500 labelled items. On `devtest` – which is not used as a training split, but is used for per-epoch checkpoint selection (Section 5) – the 32B is clearly the best single model in both languages (Table 1). On the blind test set this advantage does not transfer for English: a 32B-only ensemble scores worse than a 7B-only ensemble, despite the 32B's much better `devtest` numbers. We also trained a 32B adapter on `train` only under otherwise identical settings; it matches the `train+dev` 32B on `devtest` but does not close the English test gap either, so the extra 500 development items are not the sole explanation.

Ensembling. We average softmax-normalised per-item scores across adapters and pick the argmax, following the recipe from our SHROOM-Visions system (Yamin et al., 2026). We ensemble (i) several 7B seeds, (ii) several 32B seeds/variants, and (iii) a weighted cross-family mix. **Exact composition of the final systems.** English: five train-only 7B seeds plus one 32B adapter, the higher-resolution (1024-token) `train+dev` 32B, all at equal weight. MSA: five `train+dev` 32B adapters (four standard seeds plus the higher-resolution one) at weight 3, plus seven train-only 7B seeds at weight 1. The `train`-only 32B probe of Section 6 is used only for the controlled comparison, not as a final ensemble member. **Weight selection.** The only tuned knob is the family weight ratio; we searched a small grid (7B : 32B $\in \{1{:}1, 3{:}1, 1{:}3\}$, plus which single 32B member to include for English) on `devtest`, then confirmed the choice on the blind leaderboard. **Best-of-N disclosure.** The test phase allowed 16 submissions per track (best counts, board hidden), so our reported 0.033 / 0.056 are best-of-N; Table 4 (Appendix C) lists every scored submission per track. The gain from the first fine-tuned ensemble (0.038 / 0.072) to the final blend tracks configuration changes, not resubmission of one system.

5 Experimental Setup

We use `Qwen2.5-VL-7B-Instruct` and `Qwen2.5-VL-32B-Instruct`; we reused this backbone from our SHROOM-Visions system (Yamin et al., 2026), where it was already validated and cached, rather than the contemporaneous Qwen3-VL, to keep the two shared tasks on a common, reproducible stack. **Prompt language.** Our instruction wrapper (Appendix A) is in English for both tracks; only the three MSA statements are in Arabic (we reused the SHROOM backbone and prompt). We tested whether this matters: re-running the 7B zero-shot MSA `devtest` evaluation with an all-Arabic instruction gave a small improvement (CI 0.102 vs. 0.112, single run), so an Arabic wrapper may help slightly. Our fine-tuned systems were trained with the English wrapper, so exploiting this would need re-fine-tuning – a concrete future direction, and one that runs counter to text-only findings that English prompts are competitive for Arabic (Kmainasi et al., 2024), suggesting the effect is setting-dependent. We apply *no* Arabic-specific preprocessing – no Alif or diacritic normalisation and no segmentation – feeding the provided statements directly to the model's own tokeniser, so the pipeline is identical across languages apart from the statement text. Models are loaded in bf16 (7B) or 4-bit NF4 (32B, QLoRA) with the HuggingFace Transformers and PEFT libraries (Wolf et al., 2020; Mangrulkar et al., 2022). LoRA optimisation uses AdamW, learning rate 1×10^{-4}, 3% linear warm-up, batch size 1–2 with gradient accumulation 8 (7B) or 16 (32B), bf16 autocast; three epochs for 7B, two for 32B. Each epoch we keep the checkpoint with the best mean selection accuracy over English and MSA on a held-out split: `dev` for our initial 7B seeds, and `devtest` for all later adapters (including every 32B) once its labels were released. No fine-tuned model is trained on its own selection split, but because that split is used to pick the epoch, the corresponding validation figures are mildly optimistic. Images are processed at up to 512 visual tokens (7B/32B) or 1024 (one higher-resolution 32B probe, Appendix A), via `qwen_vl_utils`. We treat the blind `test` leaderboard as the final arbiter between configurations that look comparable on `devtest`, since we found `devtest` alone to be misleading *across* model families (Section 6).

	English		MSA	
Configuration	Devtest	Test	Devtest	Test
3B baseline (organisers)	.313	–	.490	–
7B zero-shot	–	.058	–	.110
7B FT, 1 seed (train+dev)	.024	–	.060	–
32B FT, 1 seed (train)	.022	–	.040	–
32B FT, 1 seed (train+dev)	.018	–	.038	–
7B FT, ens. (train)	.026	.038	.058	.072
32B FT, ens. (train+dev)	.020	.049	.042	.066
7B+32B ensemble	–	.034	–	.058
+ hi-res 32B (final)	–	**.033**	–	**.056**

Table 1: Selection CI (lower is better) on the labelled `devtest` split (500 items) and the blind `test` leaderboard (1,000 items). The 3B row is the organisers' reported baseline; our zero-shot models were validated on the earlier `dev` split (numbers in the text), as `devtest` labels were released only mid-phase. The single-seed block partly controls the 7B-vs-32B comparison: at matched `train+dev` data the 32B still beats the 7B on `devtest`, and `train`-only vs `train+dev` 32B are nearly identical, so the extra 500 dev items are not the driver. One confound we do *not* isolate is precision (7B in bf16 vs 32B in 4-bit NF4); see Limitations. Adapters selected on `devtest` are never trained on it (Section 5), but those `devtest` figures can be mildly optimistic. Dashes mark splits we did not measure. Final submitted system in bold.

6 Results and Analysis

Table 1 (plotted in Figure 2, Appendix B) summarises our progression. Fine-tuning is the single largest lever: on the blind test it moves English CI from a zero-shot 0.058 to 0.033–0.038 for fine-tuned ensembles, and MSA from 0.110 to 0.056–0.072. The value of same-family seed-ensembling is mixed: on the blind test it reduces variance and helps (the 7B ensemble at 0.038 beats a single 7B seed at 0.045 in English), but on `devtest` a single strong seed can match or beat a small ensemble (32B: 0.018 single vs 0.020 four-seed), so we do not claim seed-ensembling is uniformly beneficial – its main role is variance reduction on the blind set.

The result we consider most useful to report is the cross-family generalisation gap. On `devtest`, the 32B configurations consistently outperform the 7B configurations by a wide margin in both languages (Table 1). On the blind `test` set, a pure 32B ensemble is *worse* than a pure 7B ensemble in English (0.049 vs. 0.038) and only marginally better in MSA (0.066 vs. 0.072); an English-only ensemble of seven 7B seeds also failed to beat five, plateauing at 0.038, so more

members of the weaker family did not close the gap either. `devtest` was released with labels partway through the test phase for local validation. For the *train*-only models (the 7B seeds and the train-only 32B), `devtest` is genuinely held out and its rankings matched the labelled `dev` split; we do *not* use `dev` to validate the `train+dev` 32B, since `dev` is part of its training data. `devtest` misled us only *across* families. We hypothesise that the higher-capacity 32B adapters are more sensitive to their training distribution than the 7B adapters – a hypothesis supported by our `train`-only 32B, which stays strong on `devtest` (Table 1) yet still fails to close the English test gap. We cannot fully separate this from a precision effect, since the 32B runs in 4-bit while the 7B runs in bf16 (Limitations).

Despite this, our best configuration in both languages is a *cross-family* mixture (five 7B seeds plus one 32B for English, CI 0.034; four 32B at 3× the weight of five 7B for MSA, CI 0.058): replacing it with either pure-family ensemble is worse, so the 32B contributes complementary errors even though it generalises worse alone. It is thus worth keeping a higher-capacity model as a minority ensemble member rather than trusting or discarding it. Adding a higher-resolution 32B (Appendix A) improved both languages to our final CI of 0.033 / 0.056, though its own `devtest` score was no better than the other 32B adapters – diversity of errors, not single-member strength, is what the ensemble trades on.

Quantifying the complementarity. To check that this is real diversity and not noise, we compare the pure 7B-family and 32B-family ensembles item-by-item against the released blind-test gold. In English the two families agree on 94.4% of the 1,000 items; on the 56 disagreements the 7B is right 32 times and the 32B is uniquely right on 21 items (2.1% of the set) that the 7B gets wrong, with neither right on only 3. So the 32B, though weaker overall (CI 0.049 vs. 0.038), still contributes correct answers the 7B misses. An oracle that always trusted the right family would reach CI 0.017, well below either alone. MSA shows the same pattern (93.0% agreement; of 70 disagreements the 32B is uniquely right 36 times vs. the 7B's 30; oracle CI 0.036) – direct evidence the two scales make different errors.

MSA remains harder than English throughout, consistent with the released dev-phase leaderboard, where the best teams reported CI around 0.028 (English) and 0.036 (MSA) on `devtest` – MSA's gap to that reference is larger than English's at every stage of our pipeline, which we attribute to Qwen2.5-VL's comparatively weaker Arabic grounding rather than to our selection or ensembling method, since the same recipe closes most of the English gap.

6.1 Error Analysis

The official test-set gold labels and item metadata (country, category, subcategory) were released after the leaderboard closed (Abdaljalil et al., 2026); this analysis is therefore *post-submission*, on our exact final predictions (it reproduces the leaderboard's combined accuracy exactly: 96.70% English, 94.40% MSA). A three-way confusion matrix of gold vs. predicted grounded index is close to diagonal with no bias toward any statement position, as expected since the grounded index is randomised and our joint prompt is symmetric. Breaking accuracy down by *category*, the hardest category is the *same in both languages*: Food & Cooking (92.5% English, 90.3% MSA, vs. 96.7%/94.4% overall), while History, Geography & National Identity is near ceiling in both (100.0%/98.4%) – a genre effect, not a language effect: fine-grained dish or ingredient distinctions are visually harder than broad landmarks. By *country*, Bahrain is hardest in both tracks (90.8%/91.8%), with smaller Gulf countries generally below average, consistent with our aggregate finding that Qwen2.5-VL's grounding is weaker for MSA. Full tables and representative examples are in Appendix B.

7 Conclusion

Reading the metric's exact definition collapsed the task into three-way selection, for which any scorer that ranks the three statements well is a strong baseline; low-rank fine-tuning of Qwen2.5-VL on this objective was our largest single gain. Our main finding is that the model that looks best on offline validation need not be the best ensemble member on a blind test: a 32B adapter that generalised worse alone than the 7B in English still improved the English ensemble as a minority member. We read this as a caution against picking a single "best" checkpoint from offline validation when the real target is a hidden test set.

Limitations

We report a single training run per configuration rather than averaging over many seeds per configuration, so some of the differences in Table 1 sit within a few items of leaderboard noise (1,000 test items). We participated in Task 1b (English, MSA) only. Two comparisons are only partly controlled: (i) we compared English and all-Arabic MSA instructions in *zero-shot* only (a small gain for Arabic, CI 0.102 vs. 0.112); whether it survives fine-tuning is untested, as our systems were trained with the English wrapper; and (ii) the 7B runs in bf16 while the 32B runs in 4-bit NF4, so scale and precision are confounded in the 7B-vs-32B comparison – a bf16 32B or a 4-bit 7B would separate them, which we could not run within our compute budget. The error analysis (Section 6.1) uses gold labels released after the leaderboard closed, so it is post-submission, not part of our official run.

Ethical Considerations

We use only the data released by the shared task and add no new annotations or personal data. Our system detects, rather than generates, hallucinated content, and is not a deployed safety filter: its error rate is well above zero, and higher on MSA than English. We fine-tune existing open pretrained models and inherit whatever biases they carry, likely more pronounced for Arabic given the weaker zero-shot MSA performance observed. Performance is also uneven across the represented countries – our post-hoc analysis (Appendix B) finds the system weakest on Bahrain and several smaller Gulf settings – so it should not be assumed to work equally well for every country or community in the data. Training used on the order of a few GPU-days across four 80 GB GPUs.

Acknowledgments

We thank the ImageEval 2026 organisers for the extended test-phase deadline and the mid-competition release of labelled `devtest` data, both of which materially improved our ability to validate configurations before the final leaderboard closed.

References

Samir Abdaljalil, Hunzalah Hassan Bhatti, Ahlam Bashiti, Farina Amir, Md Arid Hasan, Basel Mousi, Nadir Durrani, Fahim Dalvi, Zien Sheikh Ali, Erchin Serpedin, Hasan Kurban, Mustafa Jarrar, Shammur Absar Chowdhury, and Firoj Alam. 2026. ImageEval 2026: Culturally grounded Arabic multimodal evaluation. In *Proceedings of the Fourth Arabic Natural Language Processing Conference: Shared Tasks*, Budapest, Hungary. Association for Computational Linguistics.

Firoj Alam, Ali Ezzat Shahroor, Md. Arid Hasan, Zien Sheikh Ali, Hunzalah Hassan Bhatti, Mohamed Bayan Kmainasi, Shammur Absar Chowdhury, Basel Mousi, Fahim Dalvi, Nadir Durrani, and Natasa Milic-Frayling. 2025. OASIS: A multilingual and multimodal dataset for culturally grounded spoken visual QA. *arXiv preprint arXiv:2510.06371*.

Tim Dettmers, Artidoro Pagnoni, Ari Holtzman, and Luke Zettlemoyer. 2023. QLoRA: Efficient finetuning of quantized LLMs. *Advances in Neural Information Processing Systems*.

Edward J Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. 2022. LoRA: Low-rank adaptation of large language models. In *Proceedings of ICLR*.

Mohamed Bayan Kmainasi, Rakif Khan, Ali Ezzat Shahroor, Boushra Bendou, Maram Hasanain, and Firoj Alam. 2024. Native vs non-native language prompting: A comparative analysis. *arXiv preprint arXiv:2409.07054*.

Sourab Mangrulkar, Sylvain Gugger, Lysandre Debut, Younes Belkada, Sayak Paul, and Benjamin Bossan. 2022. PEFT: State-of-the-art parameter-efficient fine-tuning methods.

Basel Mousi, Fahim Dalvi, Shammur Chowdhury, Firoj Alam, and Nadir Durrani. 2026a. Said aloud, read different: Cross-modal instability in multimodal models. In *Proceedings of Interspeech 2026*, Sydney, Australia. Accepted.

Basel Mousi, Fahim Dalvi, Shammur Absar Chowdhury, Firoj Alam, and Nadir Durrani. 2026b. Once correct, still wrong: Counterfactual hallucination in multilingual vision-language models. In *Findings of the Association for Computational Linguistics: ACL 2026*, pages 4763–4788, San Diego, California, United States. Association for Computational Linguistics.

Qwen Team. 2025. Qwen2.5-VL technical report. *arXiv preprint arXiv:2502.13923*.

Raúl Vázquez, Aman Sinha, Chuyuan Li, Artem Shelmanov, Artem Vazhentsev, Claudio Savelli, Eduardo Calò, Emilio Raimond, Stella Frank, Hengyu Luo, Flavio Giobergia, Vincent Segonne, Lorenzo Vaiani, Jörg Tiedemann, and Timothee Mickus. 2026. Overview of SHROOM-visions 2026: A shared task on hallucination detection in large vision-language models. *Preprint*, arXiv:2608.25662.

Thomas Wolf, Lysandre Debut, Victor Sanh, et al. 2020. Transformers: State-of-the-art natural language processing. In *Proceedings of EMNLP: System Demonstrations*.

Muhammad Yamin, Sheikh Abdul Wahid, and Ali Arshad. 2026. Dynamos at SHROOM-vision: Metric-aligned token tagging and ensembles for multilingual vision-language hallucination-span detection. System description paper, SHROOM-Visions 2026 shared task (UncertaiNLP workshop, EMNLP 2026).

A Training and Inference Details

LoRA/QLoRA. Rank 16, $\alpha = 32$, dropout 0.05, target modules `q,k,v,o_proj` and `gate,up,down_proj`, task type causal LM; only adapter weights and (for the 32B) the 4-bit NF4 quantisation of the frozen backbone are used, so no full-precision copy of the 32B model is required. AdamW, learning rate 1×10^{-4}, 3% linear warm-up, bf16 autocast, gradient clipping at norm 1.0. 7B: batch size 2 (effective 16 with accumulation), 3 epochs, roughly 2–3 hours per epoch on one accelerator. 32B: batch size 1 (effective 16), 2 epochs, roughly 4 hours per epoch. Training examples pool English and MSA together (train-only: 3,000/language; train+dev: 3,500/language).

Higher-resolution member. One further 32B adapter was trained identically but with the image processor's pixel budget raised from 512 to 1,024 patch tokens, to test whether fine-grained cultural detail (e.g. small text, regional dress, or architectural motifs) was being lost at the lower resolution, at roughly four times the wall-clock cost per epoch. In isolation on `devtest` it was no stronger than the matched lower-resolution 32B adapters in either language, consistent with the generalisation gap discussed in Section 6. We nonetheless added it as one further ensemble member in our final submission, where it improved both languages slightly (Table 1), reinforcing that its value lies in the different errors its higher input resolution induces rather than in any standalone strength.

Prompts. Per-statement framing: *"You are checking a statement against an image for visual hallucination. Look only at what the image actually shows. Statement: {s}. If the image clearly supports the statement, answer True. If the statement describes something not in the image or contradicted by it, answer False. Answer with only one word: True or False."* Joint framing: *"Look carefully at the image. Exactly one of the following statements is grounded in the image; the other two are not supported by it. 0) {s0} 1) {s1} 2) {s2} Which statement is grounded in the image? Answer with only the number 0, 1, or 2."*

Ensembling. Per-model scores are softmax-normalised per item before weighted averaging, so that models with different logit scales (e.g. 4-bit vs. bf16) are comparable; the final prediction is the argmax of the averaged distribution.

B Error Analysis: Additional Results

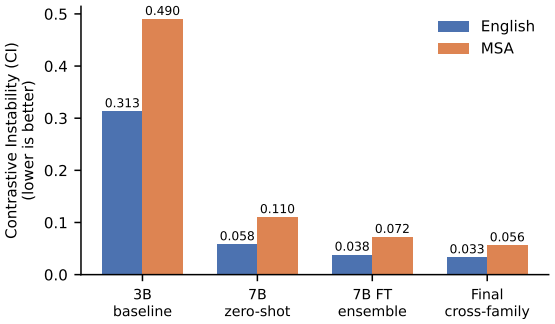


Figure 2: CI across our progression (Table 1). The 3B baseline is the organisers' reported score; the other bars are our blind-test CI. Fine-tuning gives the largest single drop; the final cross-family ensemble adds a further, smaller gain in both languages.

Table 2 and Table 3 give the full per-category and per-country accuracy of our final submitted predictions against the gold test labels released after the leaderboard closed (item counts are identical across languages, since both tracks share the same 1,000 images). Figure 1's scoring pipeline is applied identically regardless of category or country; the spread below reflects the underlying vision-language model's grounding, not our selection method.

Representative errors. In English (Bahrain, *Objects, Materials & Clothing*): given an image of boats, our system picked *"The boats visible in the image are tourist boats"* over the gold *"The boats visible in the image are fishing boats"* (the distractor *"cargo ships"* was correctly rejected) – a fine-grained visual distinction between boat types that the model's grounding could not resolve. In English (Palestine, *Vehicles & Transportation*): given an image of a building, our system picked *"The building shown in the image is a religious temple"* over the gold *"... is a public transport station"*, suggesting the model over-weighted the building's ornate facade over functional cues.

Category	n	Acc. EN	Acc. MSA
Food & Cooking	93	92.5	90.3
People, Society & Education	185	96.8	93.0
Sports & Recreation	101	97.0	93.1
Culture, Arts & Entertainment	175	96.6	94.3
Geography, Buildings & Landmarks	53	96.2	94.3
Religion & Spirituality	87	98.9	94.3
Objects, Materials & Clothing	94	95.7	96.8
Vehicles & Transportation	148	97.3	96.6
History, Geography & National Identity	64	100.0	98.4

Table 2: Selection accuracy (%) by item category, both languages, on our final submitted test predictions. Food & Cooking is the hardest category in both languages; History/Geography/National Identity the easiest.

Country	n	Acc. EN	Acc. MSA
Bahrain	98	90.8	91.8
Saudi Arabia	28	96.4	89.3
Palestine	84	97.6	92.9
Kuwait	91	96.7	93.4
Yemen	78	94.9	94.9
Iraq	85	96.5	94.1
Lebanon	76	96.1	94.7
Sudan	80	98.8	93.8
Syria	83	97.6	95.2
Morocco	85	97.6	95.3
Algeria	87	97.7	96.6
Tunisia	44	97.7	97.7
UAE	81	100.0	96.3

Table 3: Selection accuracy (%) by country, both languages, on our final submitted test predictions. Bahrain is the hardest country in both languages.

In MSA (Kuwait, *People, Society & Education*): given an image of a green flag with an emblem, our system picked the statement translating to *"… represents the Arab League"* over the gold *"… represents the Gulf Cooperation Council"*: both are green, Arabic-script regional-body flags, distinguished only by a small emblem rather than the flag's overall appearance. In MSA (Bahrain, *Religion & Spirituality*): given an image of a woman, our system picked *"… is likely performing religious rituals"* over the gold *"… is likely admiring the artwork and design"*, again a case where a plausible but visually unsupported cultural inference was preferred over the correct, more mundane reading. All four examples fall in categories or countries from Tables 2–3 that are below the respective language's average accuracy.

C Scored Test Submissions

For transparency about the best-of-N result, Table 4 lists our scored blind-test submissions per track and their official CI. Both tracks allowed 16 submissions (best counts, leaderboard hidden until close). The progression is driven by configuration changes, not by resubmitting the same system; the final rows are the submitted systems.

Track	Submitted configuration	CI ↓
EN	7B zero-shot (safety net)	.058
	single 7B seed	.045
	4×32B ensemble	.049
	3–5×7B ensemble	.038
	5×7B + 1×32B	.034
	5×7B + hi-res 32B (final)	**.033**
MSA	7B zero-shot (safety net)	.110
	single 7B seed	.077
	3×7B ensemble	.072
	4×32B ensemble	.066
	4×32B(w3) + 5×7B	.058
	5×32B(w3) + 7×7B + hi-res (final)	**.056**

Table 4: Representative scored test submissions per track (lower CI is better), showing the progression to the final best-of-N systems (bold).